Further Mathematics · Core Pure 2

Chapter 1

Complex numbers

Exponential form · De Moivre's theorem · Trigonometric identities · Series · Roots

1. Exponential form

Write $z = re^{i\theta}$.

2. Products and quotients

Multiply moduli and add arguments; divide moduli and subtract arguments.

3. De Moivre's theorem

Find powers of complex numbers.

4. Trigonometric identities

Connect multiple angles and powers of sine and cosine.

5. Complex series

Use geometric series and compare real and imaginary parts.

6. Roots

Find roots and interpret regular polygons on Argand diagrams.

RECAP: Modulus-Argument Form — Luisa · 3

For $z = x + iy$, let $r = |z|$ and $\theta = \arg z$.

Find expressions for r, x, y.

Coordinates

Modulus–argument form

$z = r(\cos\theta + i\sin\theta)$.

The principal argument satisfies $-\pi < \theta \leq \pi$.

Use the quadrant as well as $\tan\theta = y/x$. In the first quadrant, $\theta = \tan^{-1}(y/x)$.

The argument of zero is undefined.

RECAP: Modulus-Argument Form — Luisa · 4

Write -1 in modulus–argument form.

a

Write i in modulus–argument form.

b

(c) $1 + i$

c

(d) $-\sqrt{3} + i$

d

Exponential Form — Luisa · 5

Euler's formula

Recall the Taylor expansions:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Substitute $x = i\theta$ into the exponential series.

Substitute

Group

Identify

Exponential Form — Luisa · 6

Convert to exponential form

(a) -1

a

(b) $2 - 3i$

b

(c) $\sqrt{2}(\cos\frac{\pi}{10} + i\sin\frac{\pi}{10})$

c

(d) $-1 + i$

d

(e) $2(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5})$

e

Arguments differing by $2k\pi$ describe the same number. Choose the principal argument when requested.

A Final Example

Luisa · 7

Show that

$\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$.

Answer

Exercise 1A - Core Pure Year 2 Page 5

Luisa · 8

Exercise 1A · Question 1

Express in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$. Use exact values where possible, otherwise 3 significant figures.

(a) -3

a

(b) $6i$

b

(c) $-2\sqrt{3} - 2i$

c

Exercise 1A · Question 2

Express in the form $x + iy$, where $x, y \in \mathbb{R}$.

(a) $e^{i\pi/3}$

a

(b) $4e^{i\pi}$

b

(c) $3\sqrt{2}e^{i\pi/4}$

c

(d) $8e^{i\pi/6}$

d

(e) $3e^{-i\pi/2}$

e

(f) $e^{5i\pi/6}$

f

Exercise 1A · Question 3

Express as $r(\cos\theta + i\sin\theta)$, with $-\pi < \theta \leq \pi$.

(a) $e^{16i\pi/13}$

a

(b) $4e^{17i\pi/5}$

b

(c) $5e^{-9i\pi/8}$

c

Multiplying and Dividing Complex Numbers

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$.

Multiply

Product

Trigonometric product

Divide · $z_2 \neq 0$

Quotient

Trigonometric quotient

Examples

1. Multiply

$3(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12})$
$\times 4(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12})$.

1

2. Multiply

$2(\cos\frac{\pi}{15} + i\sin\frac{\pi}{15})$
$\times 3(\cos\frac{2\pi}{5} - i\sin\frac{2\pi}{5})$.

2

3. Write in exponential form

$$\frac{2(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12})}{\sqrt{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})}.$$

3

Tip

$\cos\theta - i\sin\theta = \cos(-\theta) + i\sin(-\theta)$.

Test Your Understanding — Luisa · 13

Edexcel FP2 June 2013 · Q2

$z = 5\sqrt{3} - 5i$. Find (a) $|z|$ (1 mark); (b) $\arg z$, in terms of π (2 marks).

a

b

$w = 2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$.

Find (c) $|w/z|$ (1 mark); (d) $\arg(w/z)$ (2 marks).

c

d

Exercise 1B - Core Pure Year 2 Pages 7-8 — Luisa · 14

Exercise 1B · Question 6

Use exponential form to show that

$$\frac{(\cos 9\theta + i\sin 9\theta)(\cos 4\theta + i\sin 4\theta)}{\cos 7\theta + i\sin 7\theta}$$

$\equiv \cos 6\theta + i\sin 6\theta$.

Answer

Exercise 1B · Question 7

$z = 1 + i\sqrt{3}$, $\mathrm{Re}(z^2/w) = 0$ and $|z^2/w| = |z|$.

Use geometrical reasoning to find both possibilities for w in exponential form. (4 marks)

Answer

Exercise 1B · Question 8

(a) Evaluate $(1 + i)^2$ in exponential form.

a

(b) Prove by induction that

$(1 + i)^n = 2^{n/2} e^{in\pi/4}, \quad n \in \mathbb{Z}^+$.

b

(c) Hence find $(1 + i)^{16}$.

c

Exercise 1B · Question 9

Use Euler's relations for $e^{i\theta}$ and $e^{-i\theta}$ to verify $\cos^2\theta + \sin^2\theta \equiv 1$.

Answer

Challenge 1B

(a) Prove by induction, for positive integers n, $(re^{i\theta})^n = r^n e^{in\theta}$.

Use only the complex multiplication rule, not an assumed law of indices.

a

(b) Given $z^{-n} = 1/z^n$, show that $(re^{i\theta})^{-n} = r^{-n}e^{-in\theta}$ for $r > 0$. Use the complex division rule.

b

De Moivre's Theorem

De Moivre's theorem

If $z = r(\cos\theta + i\sin\theta)$, what is z^n?

Theorem

Prove the result by induction for positive integers n.

Edexcel FP2 June 2013 · Q4

Induction

Alternative

De Moivre's Theorem for Exponential Form

If $z = re^{i\theta}$, find z^n.

Answer

Examples

1. Simplify

$$\frac{(\cos\frac{9\pi}{17}+i\sin\frac{9\pi}{17})^5}{(\cos\frac{2\pi}{17}-i\sin\frac{2\pi}{17})^3}.$$

1

2. Express in the form $x+iy$

$(1+\sqrt{3}i)^7$.

2

Test Your Understanding

Edexcel FP2 June 2010 · Q4

$z=-8+8\sqrt{3}i$.

(a) Find the modulus and argument of z. (3 marks)

(b) Use De Moivre's theorem to find z^3. (2 marks)

a

b

Exercise 1C · Question 5

Express $(3+i\sqrt{3})^5$ as $a+bi\sqrt{3}$, where a, b are integers.

Answer

Exercise 1C · Question 6

$w = 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$.

Find w^4 exactly in the form $a+ib$.

6

Question 7

$z = \sqrt{3}(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4})$.

Find z^6 exactly in the form $a+ib$.

7

Exercise 1C · Question 8

(a) Express

$$\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$$

as $re^{i\theta}$, with $r > 0$ and $-\pi < \theta \leq \pi$.

a

(b) Find the smallest positive integer n for which this number raised to the power n is real and positive.

b

Exercise 1C · Question 9

Use De Moivre's theorem to show that

$(a+bi)^n + (a-bi)^n$

is real for integer n, wherever the powers are defined.

Answer

Challenge 1C

Without Euler's relation, prove for positive integers n and $r > 0$:

$[r(\cos\theta + i\sin\theta)]^{-n}$

$= r^{-n}[\cos(-n\theta) + i\sin(-n\theta)]$.

You may assume De Moivre's theorem for positive integers.

Answer

Express $\cos 3\theta$ in powers of $\cos\theta$.

How can De Moivre produce a $\cos 3\theta$ term?

De Moivre

Now expand the left side. Write $c = \cos\theta$, $s = \sin\theta$.

Expand

Compare the two expressions.

Real parts

Applications of de Moivre #1: Trig identities

Luisa · 29

Use De Moivre to express (a) $\cos 6\theta$, (b) $\sin 6\theta / \sin\theta$, in powers of $\cos\theta$. For (b), $\theta \neq k\pi$.

Set $c = \cos\theta$, $s = \sin\theta$.

$$(c+is)^6 = c^6 + 6ic^5 s - 15c^4 s^2 - 20ic^3 s^3 + 15c^2 s^4 + 6ics^5 - s^6.$$

a

b

Test Your Understanding

Luisa · 30

Edexcel FP2 June 2011 · Q7

Use De Moivre's theorem to show

$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$.

(5 marks)

Given $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, solve $\sin 5\theta = 5\sin 3\theta$ for $0 \leq \theta < 2\pi$, to 3 decimal places. (6 marks)

Identity

Solutions

Identities for powers of sine and cosine — Luisa · 31

Can we reverse the process: express powers of sine and cosine using multiple angles?

Let $z = \cos\theta + i\sin\theta$. Find $z + z^{-1}$ and $z - z^{-1}$.

First powers

Use De Moivre to generalise.

General powers

These results can be derived when needed. Expand powers of $z + z^{-1}$ or $z - z^{-1}$, then group matching exponents.

Identities for powers of sine and cosine — Luisa · 32

Express $\cos^5\theta$ in terms of cosines of multiples of θ.

Let $z = e^{i\theta}$, so $z + z^{-1} = 2\cos\theta$.

Group

Start

Substitute

Expand

The exponents descend in steps of 2. Pair equal positive and negative powers.

Identities for powers of sine and cosine — Luisa · 33

Express $\sin^3\theta$ in terms of sines of multiples of θ.

Use $z - z^{-1} = 2i\sin\theta$.

Start

Expand

Group

Simplify

The binomial signs alternate. Dividing by $-8i$ changes the signs.

Test Your Understanding — Luisa · 34

(a) Express $\sin^4\theta$ in terms of cosines of multiples of θ.

(b) Hence evaluate $\int_0^{\pi/2} \sin^4\theta\, d\theta$.

Expand

a

b

Exercise 1D · Question 2

(a) Use De Moivre's theorem to show

$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$.

(b) Given $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, solve $\cos 5\theta + 5\cos 3\theta = 0$ for $0 \leq \theta < \pi$, to 3 decimal places.

a

b

Exercise 1D · Question 3

(a) Prove

$32\cos^6\theta = \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$.

(6 marks)

(b) Evaluate $\int_0^{\pi/6} \cos^6\theta \, d\theta$. (3 marks)

a

b

Exercise 1D · Question 4

(a) Prove

$32\cos^2\theta\sin^4\theta = \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2.$

(b) Evaluate $\int_0^{\pi/3} \cos^2\theta\sin^4\theta\, d\theta$.

a

b

Exercise 1D · Question 6

(a) Show

$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1.$

(b) Find the six roots of

$32x^6 - 48x^4 + 18x^2 - \frac{3}{2} = 0$

to 3 decimal places.

a

b

Exercise 1D · Question 7

(a) Show $\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$. (4 marks)

(b) Show

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}.$$

(4 marks)

(c) Solve $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$, to 2 decimal places. (5 marks)

a

b

c

Sums of Series **Luisa · 40**

Complex geometric series

A geometric series starts $w, wz, wz^2, \ldots$. The common ratio is z.

Finite sum

Convergence

The modulus measures distance from zero: $|-4| = 4$, $|3 + 4i| = 5$.

Infinite sum

Example

Luisa · 41

For a positive integer n, let $z = e^{i\pi/n}$. Prove

$1 + z + \cdots + z^{n-1} = 1 + i \cot \dfrac{\pi}{2n}$.

Answer

Let's practise that hard bit...

Luisa · 42

Half-angle rearrangements

Factor out the exponential with half the angle:

$e^{2i\alpha} - 1 = 2ie^{i\alpha} \sin \alpha$.

Simplify (1) $\dfrac{3}{e^{2i\theta} - 1}$.

1

(2) $\dfrac{4e^{i\theta}}{e^{4i\theta} - 1}$

2

(3) $\dfrac{1}{e^{i\theta/3} - 1}$

3

These formulas apply where the original denominators are non-zero.

Using mod-arg form to split summation

Using modulus–argument form to split a sum

For $\theta \neq 2k\pi$,

$S = e^{i\theta} + e^{2i\theta} + \cdots + e^{ni\theta}$

$= \dfrac{e^{i\theta}(e^{ni\theta} - 1)}{e^{i\theta} - 1}.$

The first term is $e^{i\theta}$, common ratio $e^{i\theta}$, and there are n terms.

Expand each exponential using Euler's formula:

$S = (\cos\theta + i\sin\theta) + \cdots$

$+ (\cos n\theta + i\sin n\theta).$

Group the real and imaginary parts:

$\operatorname{Re} S = \cos\theta + \cdots + \cos n\theta,$

$\operatorname{Im} S = \sin\theta + \cdots + \sin n\theta.$

Thus one complex geometric sum gives both trigonometric sums.

Example

Let

$P = \cos\theta + \cos 2\theta + \cdots + \cos 8\theta,$

$Q = \sin\theta + \sin 2\theta + \cdots + \sin 8\theta.$

(a) Show

$P + iQ = e^{9i\theta/2}\dfrac{\sin 4\theta}{\sin(\theta/2)}.$

Assume $\sin(\theta/2) \neq 0$.

Series

Simplify

Example

For the sums P, Q on the previous slide,

$P + iQ = e^{9i\theta/2}\dfrac{\sin 4\theta}{\sin(\theta/2)}$.

(b) Find expressions for P and Q, and simplify Q/P.

P

Q

Ratio

Exercise 1E - Core Pure Year 2 Pages 18-19

Exercise 1E · Question 4

$C = 1 + \dfrac{\cos\theta}{3} + \dfrac{\cos 2\theta}{9} + \cdots,$

$S = \dfrac{\sin\theta}{3} + \dfrac{\sin 2\theta}{9} + \cdots.$

(a) Show $C + iS = 3/(3 - e^{i\theta})$.

(b) Find real expressions for C, S.

a

b

Exercise 1E · Question 5

For $0 < \theta < \pi$,

$P = 1 + \cos\theta + \cdots + \cos 12\theta$,

$Q = \sin\theta + \cdots + \sin 12\theta$.

(a) Show

$$P + iQ = e^{6i\theta}\frac{e^{13i\theta/2} - e^{-13i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}}.$$

(4 marks)

(b) Deduce real expressions for P, Q. (4 marks)

a

b

Exercise 1E · Question 6

$$C = \sum_{k=0}^{n}\binom{n}{k}\cos k\theta, \qquad S = \sum_{k=1}^{n}\binom{n}{k}\sin k\theta.$$

(a) Show $C = (2\cos(\theta/2))^n \cos(n\theta/2)$.

(b) Show $S/C = \tan(n\theta/2)$, where defined.

a

b

Exercise 1E · Question 7

(a) Show $(2+e^{i\theta})(2+e^{-i\theta}) = 5+4\cos\theta$. (2 marks)

$$C = 1 - \frac{\cos\theta}{2} + \frac{\cos 2\theta}{4} - \cdots,$$

$$S = \frac{\sin\theta}{2} - \frac{\sin 2\theta}{4} + \cdots.$$

(b) By considering $C - iS$, find C, S. (4 marks)

a

b

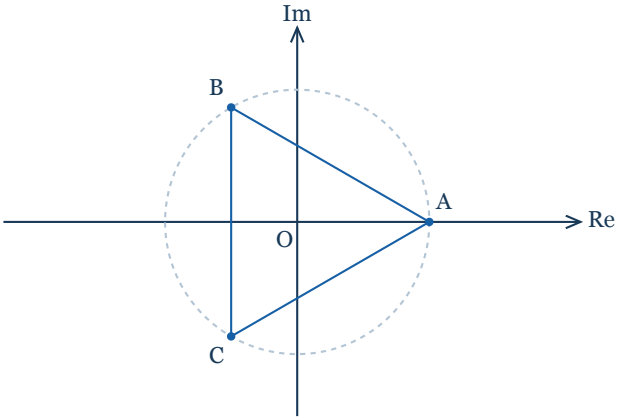
Applications of de Moivre #2: Roots **Luisa · 50**

Cube roots of unity

Solve $z^3 = 1$. Write $z = re^{i\theta}$.

Modulus and argument

Roots



The roots lie on the unit circle. Consecutive arguments differ by $2\pi/3$, so they form an equilateral triangle.

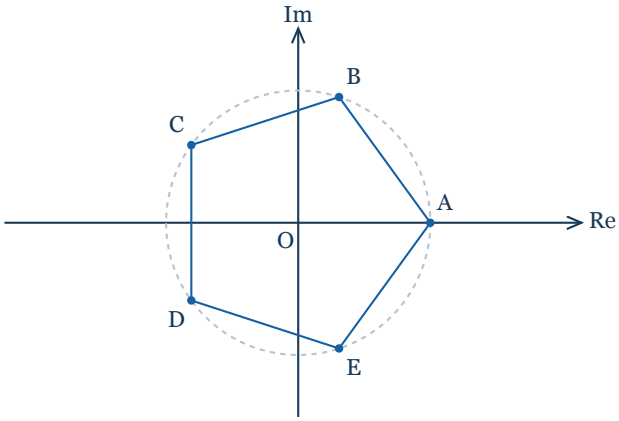
More on Roots of Unity

The nth roots of unity

$z^n = 1 \implies z = e^{2k\pi i/n}, \quad k = 0, \ldots, n-1.$

They are equally spaced on the unit circle, with angular separation $2\pi/n$.

Put $\omega = e^{2\pi i/n}$. The roots are $1, \omega, \ldots, \omega^{n-1}$.



For $n \geq 2$, show that their sum is zero.

Sum

Example

Find all four roots of

$z^4 = 2 + 2\sqrt{3}i$

in exponential form, using principal arguments.

Answer

Find the modulus and argument of $-2+2\sqrt{3}i$.

Hence solve $z^4=-2+2\sqrt{3}i$, using exponential form and principal arguments.

a

b

Exercise 1F · Question 4

(a) Find the three roots of $(z+1)^3=-1$, in the form $x+iy$.

(b) Plot the roots on an Argand diagram.

(c) Find the centre and radius of the circle through them.

a

Argand diagram

c

Exercise 1F · Question 6

(a) Find the modulus and argument of $-2-2i\sqrt{3}$.

(b) Solve $z^4+2+2i\sqrt{3}=0$. Give $re^{i\theta}$, $r>0$, $-\pi<\theta\leq\pi$, and illustrate the roots.

a

Argand diagram

b

Exercise 1F · Question 7

Find the four distinct roots of $z^4=2(1-i\sqrt{3})$ in exponential form, and show them on an Argand diagram. (7 marks)

Argand diagram

Answer

Exercise 1F · Question 8

$z = \sqrt{6} + i\sqrt{2}$.

(a) Find $|z|$ and $\arg z$. (2 marks)

(b) Find all w such that $w^3 = z^4$, in exponential form with principal arguments. (4 marks)

a

b

Exercise 1F · Question 9

(a) Solve

$1 + z + z^2 + \cdots + z^7 = 0$.

(b) Hence deduce that $z^2 + 1$ and $z^4 + 1$ are factors of this polynomial.

a

b

Challenge

(a) Find the six roots of $z^6 = 1$ as $e^{i\theta}$, with principal arguments.

(b) Hence show that the solutions of $(z+1)^6 = z^6$ are $z = -\frac{1}{2} + \frac{i}{2}\cot\frac{k\pi}{6}, \quad k = 1, \ldots, 5.$

a

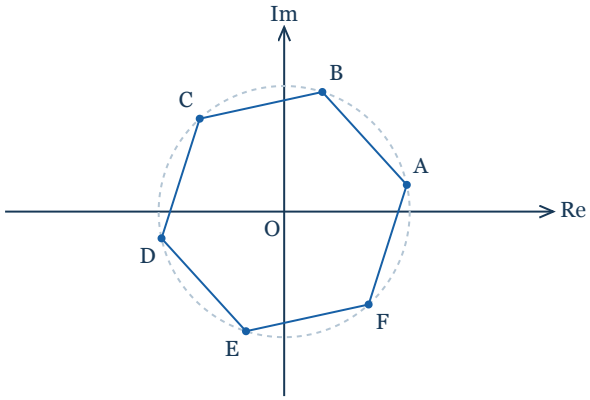
b

Roots and regular polygons

If z_1 is one root of $z^n = w \neq 0$, all roots are $z_1, z_1\omega, \ldots, z_1\omega^{n-1}, \quad \omega = e^{2\pi i/n}$.

They form a regular n-gon centred at the origin.

For example, the roots of $z^6 = 7 + 24i$ have modulus $25^{1/6}$ and angular separation $\pi/3$.



Multiplication by ω rotates a point by $2\pi/n$ without changing its distance from the origin.

Once one vertex is known, rotate repeatedly to generate the others.

Example

An equilateral triangle is centred at the origin. One vertex is $P(\sqrt{3}, 1)$.

Find the other two vertices and the area.

Vertices

Area

Argand diagram

Exercise 1G - Core Pure Year 2 Pages 26-27

Exercise 1G · Question 3

Triangle OAB is equilateral on an Argand diagram. O is the origin; A represents $\sqrt{3}(1 - i)$, and B represents b.

Find both possibilities for b in exponential form and sketch both triangles. (5 marks)

Answer

Argand diagram

Second triangle

Exercise 1G · Question 4

(a) Find the fourth roots of $-12i$, as $re^{i\theta}$ with principal arguments. Illustrate them. (6 marks)

In increasing argument order, label them A, B, C, D.

(b) Their side midpoints are the fourth roots of a complex number w. Find w. (4 marks)

a

Argand diagram

b

Exercise 1G · Question 5

A regular hexagon on an Argand diagram is centred at the origin. Vertex P represents $8 + 8i$.

Find the other five vertices in the form $a + bi$, and illustrate them.

Use $z_k = (8 + 8i)e^{ik\pi/3}$.

Argand diagram

Answer